

2023-2024
2nd year /Electronic I
Electronic and Telecommunication Department.
Engineering Science/ Kufa University

Hussain Mumtaz Hussain
hussainm.alshamy@uokufa.edu.iq

Microelectronic

Microelectronic circuit analysis and design

Donald A.Neamen 4th edition

In this section, the physical and electrical properties of the most important semiconductor devices are presented, and the use of these devices such as switches, digital applications, and amplification.

First semester

Chapter _2: Demonstrate the nonlinear characteristics of the diode in various circuits such as switching, wave shaping applications.

Chapter_3: Introduction to the first type of the semiconductor transistor metal-oxide-semiconductor field effect transistor (MOSFET) and it is DC analysis with basic application.

Chapter_4: The fundamental design of MOS transistor circuit. also, it is application as amplifier.

Chapter_5: Study the bipolar transistor including the DC analysis of bipolar transistor circuits and it is basic application as amplifier.

Second semester

Chapter_6 : Basic BJT Amplifiers

Chapter_7: Study the frequency response for both types of transistors.

Chapter_8: Design of the electronic circuit including power amplifiers and various output stages.

Introduction:

An electronic system consisting of electronic components such as power supply, signal sources, amplifiers, and digital logic circuits.

Electronics is the study of how charges move in semiconductor device such as transistor, which are the main components of modern electronic circuits.

The term "microelectronics" refers to integrated circuit (IC) technology, which creates circuits with millions of components on a single piece of semiconductor material; the microprocessor is a common example of an integrated circuit.

Aim of this study is to create and construct systems with electronic components.

Electronic circuits:

There are two inputs, the first is a DC power supply that supplies voltage and current to create the appropriate bias for the transistors. The second input is the signal, which are time-varying signals from a specific source that often need to be amplified before the signal becomes useful.

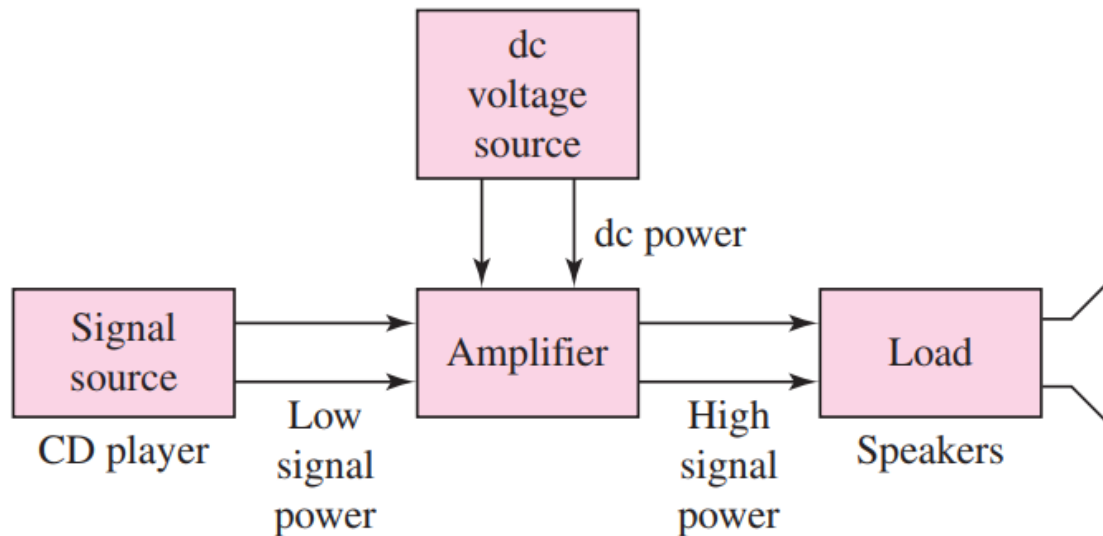


Figure 1.1 Electronic circuit with two inputs: the dc power supply input, and the signal input